Linear algebra – new problems

Problems built on recent research results

Each problem is a special case of a published result or of a lemma used in its proof; the papers are in this folder. The problems are ordered by difficulty. The difficulty is the position of the problem in an IMC day: 1 easy, 2 below average, 3 medium, 4 above average, 5 hard.

Problem 1. *IMC level: 1/5 (easy)*

For an $n \times n$ matrix $A = (a_{ij})$ with entries in $\{0,1\}$ and a vector $x \in \{0,1\}^n$ define the vector $A \odot x \in \{0,1\}^n$ by

$$(A \odot x)_i = \max_{1 \leqslant j \leqslant n} a_{ij} x_j, \qquad i = 1, \ldots, n.$$

Suppose that the map $x \mapsto A \odot x$ from $\{0,1\}^n$ to itself is injective. Prove that A is a permutation matrix. Deduce that the map $x \mapsto A^{\mathsf{T}} \odot x$ is injective as well.

Based on: S. Gu, W. Qi, Y. Cheng, *Transpose symmetry of injectivity over commutative semirings*, arXiv:2608.16205, Theorem 1 (the map $x \mapsto Ax$ is injective if and only if $x \mapsto A^{\mathsf{T}}x$ is, over every commutative semiring); here the semiring is $(\{0,1\}, \max, \min)$.

Problem 2. *IMC level: 2/5 (below average)*

Prove that for every positive integer n and every integer k with $0 \leqslant k \leqslant 2^{n-1}$ there is an $n \times n$ matrix with entries in $\{0,1\}$ whose permanent equals k.

Based on: Z. Hunter, M. Kwan, L. Sauermann, *Exponential anticoncentration of the permanent*, arXiv:2509.22577, Section 1.2 (the theorem of Brualdi and Newman).

Problem 3. *IMC level: 2/5 (below average)*

For a square matrix X let $\rho(X)$ be its spectral radius. For a matrix $A = (a_{ij})$ with positive entries and $t \in \mathbb{R}$ put $A^{(t)} = (a_{ij}^t)$.

(a) Let $A = (a_{ij})$ and $B = (b_{ij})$ be $n \times n$ matrices with nonnegative entries and $C = (\sqrt{a_{ij} b_{ij}})$. Prove that $\rho(C)^2 \leqslant \rho(A)\,\rho(B)$.

(b) Deduce that for every $n \times n$ matrix A with positive entries the function $t \mapsto \ln \rho(A^{(t)})$ is convex on $\mathbb{R}$.

Based on: K. Bogdanović, *Inequalities on the generalized and the joint spectral radius of Hadamard products*, arXiv:2311.02444, Theorem 1.2, inequality (1.8) (Kingman's theorem).

Problem 4. *IMC level: 2/5 (below average)*

Let F be a field. Call a matrix $X \in M_2(F)$ *unipotent* if $X \neq I$ and $(X - I)^2 = 0$. Let $A \in M_2(F)$ be a non-scalar matrix with $\det A = 1$. Prove that $A = XYX^{-1}Y^{-1}$ for some unipotent $X, Y \in M_2(F)$ if and only if $\operatorname{tr} A - 2 = c^2$ for some nonzero $c \in F$.

Based on: K. L. Dela Rosa, J. P. C. Santos, *On commutators of unipotent matrices of index 2*, arXiv:2409.13339, Theorem 3.2.

Problem 5. *IMC level: 2/5 (below average)*

Let $\mathbb{F}_3$ be the field with three elements. Call $D \in M_3(\mathbb{F}_3)$ *diagonalizable* if $D = P\Lambda P^{-1}$ for some invertible $P \in M_3(\mathbb{F}_3)$ and some diagonal $\Lambda \in M_3(\mathbb{F}_3)$. Let $A \in M_3(\mathbb{F}_3)$ be a matrix whose characteristic polynomial is irreducible over $\mathbb{F}_3$. Prove that $A = D+M$ for some diagonalizable $D \in M_3(\mathbb{F}_3)$ and some $M \in M_3(\mathbb{F}_3)$ with $M^2 = 0$ if and only if $\operatorname{tr} A = 0$.

Based on: P. Danchev, E. García, M. Gómez Lozano, *Matrices over finite fields of odd characteristic as sums of diagonalizable and square-zero matrices*, arXiv:2507.05762, Remark 3.2 and Corollary 3.3.

Problem 6. *IMC level: 3/5 (medium)*

Call a set $\{a, b, c, d\}$ of four distinct positive integers *spectral* if each of the 24 matrices $\begin{pmatrix} p & q \\ r & s \end{pmatrix}$, where (p, q, r, s) is a permutation of (a, b, c, d), has integer eigenvalues.

(a) Show that $\{1, 6, 7, 12\}$ is spectral.

(b) Prove that there are infinitely many spectral sets whose elements have no common divisor greater than 1.

Based on: M. J. W. Hall, *Matrices with integer eigenvalues for all permutations of coefficients (thanks to Pythagoras!)*, arXiv:2601.01083, equations (7)–(17): spectral sets obtained from Pythagorean triples.

Problem 7. *IMC level: 3/5 (medium)*

Let $A\colon \mathbb{R}^n \to \mathbb{R}^n$ be a linear operator such that $A([-1, 1]^n)$ is contained in the closed unit ball $B_1(0)$. Find the maximal possible value of $|\det A|$.

Based on: F. Grundbacher, T. Kobos, *On certain extremal Banach–Mazur distances and Ader's characterization of distance ellipsoids*, arXiv:2407.08829, Section 1 (the John and Loewner ellipsoids of the cube $[-1, 1]^n$; John's theorem).

Problem 8. *IMC level: 3/5 (medium)*

Let $k, m, r_1, \ldots, r_m$ be positive integers. The tree T is obtained from a vertex v_0 by attaching m paths: the i-th path has r_i vertices, and one of its end vertices is joined to v_0. Let B be the symmetric matrix whose rows and columns are indexed by the vertices of T, with

$$B_{v_0v_0} = k, \qquad B_{vv} = 2 \ \ (v \neq v_0), \qquad B_{uv} = \begin{cases} -1, & u, v \text{ adjacent}, \\ 0, & \text{otherwise} \end{cases} \qquad (u \neq v).$$

(a) Prove that $\det B = (r_1 + 1) \cdots (r_m + 1)\Big(k - \displaystyle\sum_{i=1}^{m} \frac{r_i}{r_i + 1}\Big)$.

(b) Prove that B is positive definite if and only if $\displaystyle\sum_{i=1}^{m} \frac{1}{r_i + 1} > m - k$.

(c) Let $k = 2$ and $m = 3$. Find all triples $r_1 \leqslant r_2 \leqslant r_3$ for which $\det B = 0$.

Based on: E. Torrente-Lujan, *Star-shaped integral Cartan-type matrices and an Egyptian-fraction classification of affine weighted trees*, arXiv:2605.23011, Proposition 4.1, Theorem 4.2 and Corollary 4.3.

Problem 9. *IMC level: 3/5 (medium)*

An *alternating sign matrix* is a square matrix with entries in $\{0, 1, -1\}$ such that in every row and in every column the nonzero entries alternate in sign, beginning and ending with $+1$ (for example, every permutation matrix is one). Prove that if A and B are $n \times n$ alternating sign matrices with $AB = I$, then A and B are permutation matrices.

Based on: C. O'Brien, R. Quinlan, *Groups of singular alternating sign matrices*, arXiv:2405.13611, Introduction (the theorem that the only invertible alternating sign matrices whose inverses are alternating sign matrices are the permutation matrices).

Problem 10. *IMC level: 3/5 (medium)*

A *tournament matrix* is an $n \times n$ matrix M with entries in $\{0, 1\}$ such that $M + M^{\mathsf{T}} = J - I$, where J is the all-ones matrix. A *cyclic triangle* is a set $\{i, j, k\}$ with $m_{ij} = m_{jk} = m_{ki} = 1$. Prove that if M is singular, then there are at most $\frac{1}{4}\binom{n}{3}$ cyclic triangles, and that for every even n this bound is attained.

Based on: M. Burnham, *Structure of singular and nonsingular tournament matrices*, arXiv:2210.12371, Lemma 2.3 (Shader's inequality) and Proposition 2.4.

Problem 11. *IMC level: 3/5 (medium)*

For a tournament on the vertex set $\{1, \ldots, n\}$ its *Seidel matrix* S is the $n \times n$ matrix with $s_{ii} = 0$, $s_{ij} = 1$ if i beats j, and $s_{ij} = -1$ if j beats i. Let n be even. Prove that every odd perfect square k^2 with $k \leqslant n - 1$ is the determinant of the Seidel matrix of some tournament on n vertices.

Based on: S. Klanderman, M. Montee, A. Piotrowski, A. Rice, B. Shader, *Determinants of Seidel tournament matrices*, arXiv:2406.09697, Question 1.1, Lemma 3.1 and Theorem 3.2.

Problem 12. *IMC level: 3/5 (medium)*

Let $x_1, \ldots, x_{2n}$ be real numbers and let A be the $2n \times 2n$ matrix with entries

$$a_{ij} = (x_i - x_j)^2 \ (i < j), \qquad a_{ii} = 0, \qquad a_{ij} = -(x_i - x_j)^2 \ (i > j).$$

Prove that $\det A = 4^{n-1} \prod_{k=1}^{2n} (x_k - x_{k+1})^2$, where $x_{2n+1} = x_1$.

Based on: A. Dzhumadil'daev, *Symmetry groups of pfaffians of symmetric matrices*, arXiv:2201.11570, Theorems 1.2 and 1.3.

Problem 13. *IMC level: 3/5 (medium)*

Let $n \geqslant 2$, let $A \in M_n(\mathbb{Q})$ be a matrix whose characteristic polynomial f is irreducible over $\mathbb{Q}$, and let $p \in \mathbb{Q}[x]$ be a nonconstant polynomial.

(a) Prove that the equation $p(X) = A$ has a solution $X \in M_n(\mathbb{Q})$ if and only if the polynomial $f(p(x))$ has a divisor of degree n in $\mathbb{Q}[x]$.

(b) Prove that the equation $X^2 = A$ has either no solutions or exactly two solutions in $M_n(\mathbb{Q})$.

Based on: G. J. Groenewald, G. Goosen, D. B. Janse van Rensburg, A. C. M. Ran, M. van Straaten, *Galois groups and rational solutions of* $p(X) = A$, arXiv:2307.06303, Theorem 4 and Proposition 10.

Problem 14. *IMC level: 3/5 (medium)*

Let $p \in \mathbb{C}[x]$ be a nonconstant polynomial. Prove that a matrix $C \in M_n(\mathbb{C})$ can be written as $C = p(AB) - p(BA)$ with $A, B \in M_n(\mathbb{C})$ if and only if $\operatorname{tr} C = 0$.

Based on: T. H. Dung, T. N. Son, P. D. Vinh, *A nonlinear analogue of additive commutators*, arXiv:2505.08268, Theorem 3.4 and Corollary 3.5.

Problem 15. *IMC level: 4/5 (above average)*

Let A be a real symmetric $n \times n$ matrix with nonnegative entries and let $x \in \mathbb{R}^n$ be a vector with nonnegative coordinates and $\|x\| = 1$. Prove that

$$x^{\mathsf{T}} A^3 x \;\geqslant\; \left(x^{\mathsf{T}} A x\right)^3.$$

Based on: C. Stoner, *The graph density domination exponent*, arXiv:2211.09870, Introduction and Section 4, Theorem 4.1 (the Blakley–Roy inequality).

Problem 16. *IMC level: 4/5 (above average)*

Let A be a complex 2×2 matrix with eigenvalues λ and μ (counted with multiplicity). Prove that the set

$$W(A) = \{\, u^* A u \;:\; u \in \mathbb{C}^2,\; u^* u = 1 \,\}$$

is the closed region bounded by an ellipse (possibly degenerating into a segment) with foci λ and μ whose minor axis has length $\sqrt{\operatorname{tr}(A^*A) - |\lambda|^2 - |\mu|^2}$.

Based on: P. Blaschke, *Coval description of the boundary of a numerical range and the secondary values of a matrix*, arXiv:2410.03744, Proposition 1 (the elliptical range theorem of Toeplitz).

Problem 17. *IMC level: 4/5 (above average)*

A real $n \times n$ matrix T is called a *transvection* if $\operatorname{rank}(T - I) = 1$ and $(T - I)^2 = 0$. For invertible matrices put $[X, Y] = XYX^{-1}Y^{-1}$.

(a) Let $n \geqslant 3$. Prove that for every invertible real $n \times n$ matrix A there are transvections T_1, T_2 with $[[A, T_1], T_2] = I$.

(b) Let $n = 2$. Prove that for an invertible real 2×2 matrix A such transvections exist if and only if A has a real eigenvalue.

Based on: I. Chinyere, *Unipotence of a double commutator with transvections*, arXiv:2609.17006, Theorem 2.9 and Remark 2.8 (the case $n = 2$ is left open there; part (b) answers it over $\mathbb{R}$).

Problem 18. *IMC level: 4/5 (above average)*

Let F be a finite field with q elements. Call a set $S \subseteq M_n(F)$ *singular* if $\det(A - B) = 0$ for all $A, B \in S$.

(a) Prove that every singular set has at most $q^{n^2 - n}$ elements, and that this bound is attained.

(b) Let $n = 2$. Prove that every singular set which is maximal with respect to inclusion has exactly q^2 elements.

Based on: S. Rahimi, A. Nikseresht, *Well-covered unitary Cayley graphs of matrix rings over finite fields and applications*, arXiv:2311.15255, Lemma 1.1 and Proposition 2.1.

Problem 19. *IMC level: 4/5 (above average)*

Let X be a finite set with $|X| = n$ and let $\mathbb{K}$ be a field. Suppose that for all $x, y, z \in X$

$$[x = y = z] = \sum_{i \in I_1} f_i(x)\, g_i(y, z) + \sum_{i \in I_2} f_i(y)\, g_i(x, z) + \sum_{i \in I_3} f_i(z)\, g_i(x, y),$$

where I_1, I_2, I_3 are disjoint finite index sets, $f_i \colon X \to \mathbb{K}$, $g_i \colon X^2 \to \mathbb{K}$, and $[x = y = z]$ equals 1 if $x = y = z$ and 0 otherwise. Prove that $|I_1| + |I_2| + |I_3| \geqslant n$.

Based on: W. T. Gowers, *The slice rank of a direct sum*, arXiv:2105.08394, Lemma 1 (Tao's lemma on the slice rank of a diagonal tensor).

Problem 20. *IMC level: 4/5 (above average)*

Let $x_1, \ldots, x_n$ ($n \geqslant 2$) be distinct points of $\mathbb{R}^d$ and let $D = \bigl(\|x_i - x_j\|\bigr)_{i,j=1}^{n}$ be the matrix of their Euclidean distances. Prove that D has exactly one positive eigenvalue and $n-1$ negative eigenvalues. In particular, D is invertible.

Based on: E. Roff, S. Willerton, *The microscopic weighting on a metric space*, arXiv:2607.05349, Section 4.1 (Euclidean spaces are of strictly negative type; Schoenberg's theorem).

Problem 21. *IMC level: 4/5 (above average)*

(a) Let G be a real symmetric positive definite $n \times n$ matrix with all diagonal entries equal to m and $|g_{ij}| \geqslant b > 0$ for all $i \neq j$. Prove that $\det G \leqslant \bigl(m + (n-1)b\bigr)(m - b)^{n-1}$.

(b) Deduce that for odd n every $n \times n$ matrix A with entries ± 1 satisfies $|\det A| \leqslant \sqrt{2n-1}\,(n-1)^{(n-1)/2}$.

Based on: P. Browne, R. Egan, F. Hegarty, P. Ó Catháin, *A survey of the Hadamard maximal determinant problem*, arXiv:2104.06756, Lemma 7, Theorem 8 (Ehlich–Wojtas) and Corollary 10 (Barba's bound).

Problem 22. *IMC level: 5/5 (hard)*

Let $n \geqslant 3$. A family $\mathcal{F}$ of distinct subsets of $\{1, \ldots, n\}$ has the following properties: every set in $\mathcal{F}$ has an odd number of elements, any two distinct sets in $\mathcal{F}$ have an odd number of common elements, and any three distinct sets in $\mathcal{F}$ have an even number of common elements. Determine the largest possible size of $\mathcal{F}$.

Based on: G. Johnston, J. O'Neill, *A few new oddtown and eventown problems*, arXiv:2312.13588, Theorem 1 (the case of odd sizes, odd pairwise and even triple intersections).

Problem 23. *IMC level: 5/5 (hard)*

Let $n \geqslant 3$ be odd. For disjoint nonempty sets $X, Y \subseteq \{1, \ldots, n\}$ let $B(X, Y)$ be the complete bipartite graph whose edges are all pairs $\{x, y\}$ with $x \in X$, $y \in Y$. Find the smallest t for which there are complete bipartite graphs $B(X_1, Y_1), \ldots, B(X_t, Y_t)$ such that every pair of distinct elements of $\{1, \ldots, n\}$ is an edge of an odd number of them.

Based on: I. Leader, T. S. Tan, *Odd covers of complete graphs and hypergraphs*, arXiv:2408.05053, Theorem 1 and Corollary 2.

Problem 24. *IMC level: 5/5 (hard)*

Let A be a complex $n \times n$ matrix. Prove that $A = B + C$ for some complex $n \times n$ matrices B and C with $B^2 = C^2 = 0$ if and only if A is similar to $-A$.

Based on: C. de Seguins Pazzis, *Sums of two square-zero selfadjoint or skew-selfadjoint endomorphisms*, arXiv:2210.03955, Theorems 1.1 and 1.2 (the theorem of Wang–Wu and Botha).

Problem 25. *IMC level: 5/5 (hard)*

Hyperplanes $H_1, \ldots, H_m$ in $\mathbb{R}^n$ together contain all 2^n points of $\{0,1\}^n$. Each H_i is given by an equation $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ in which all coefficients a_{ij} are nonzero. Prove that $m \geqslant \frac{n}{2} + 1$.

Based on: L. Sauermann, Z. Xu, *Nondegenerate hyperplane covers of the hypercube*, arXiv:2507.00773, Theorem 1.1 and Remark 2.4.